Problem Details

Community-based Influence Maximization in Attributed Networks

Core Question

Which users to incentivize so adoption spreads maximally within budget and across the right communities?

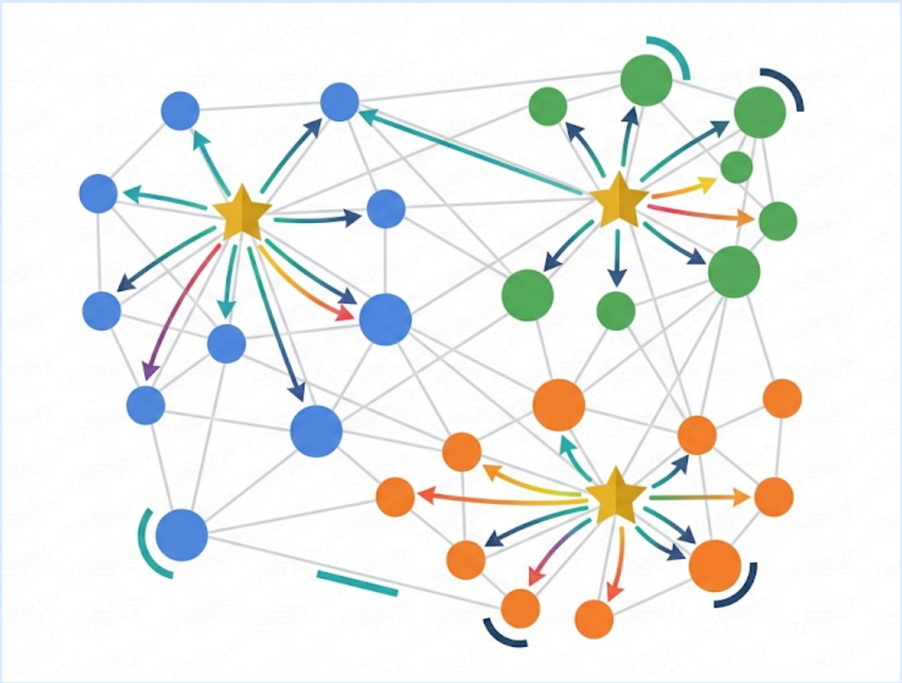

Fig: Influence propagates from seeds (★) through the network, reaching different communities

BASE PROBLEM

Influence Maximization (IM)

Select k seed nodes in a social network to maximize information spread under the Independent Cascade model. Classical greedy (CELF) and sampling-based (IMM) methods are too slow or inflexible at scale.

EXTENSION 1

Budget Constraints

Each seed node has a cost (e.g., Influencer incentive fee). Total spend must stay within a fixed campaign budget

EXTENSION 2

Group-Sensitive Objectives

Nodes belong to structural communities. The campaign must control how influence distributes across segments balanced reach across all groups, or targeted reach into specific ones.

Why is this a Good RL Problem?

Problem is sequential, budget-limited, depends on graph structure, has stochastic environment and requires reward maximisation

STATE

- **Graph topology**
  Graph structure and already influenced nodes
- **Seed set so far**
  Which nodes have already been selected as seeds in the current episode
- **Remaining budget**
  How much of the campaign spend limit is still available
- **Formally:**
  S_t = (Graph G, Selected set, Remaining Budget)

ACTION

- **Select one node**
  Select one unchosen node
- **Budget**
  Node cost must fit remaining budget
- **Large but discrete**
  Thousands of possible actions per step RL avoids exhaustive combinatorial search
- **Marginal gains**
  Sequential selection naturally captures diminishing returns of each additional seed

REWARD

- **Marginal spread**
  Reward = Marginal increase in influence spread
- **Community shaping**
  Reward term reflecting how well influence distributes across target segments
- **Total reward**
  Total episode return = Total influence spread
- **Episode Ends when Budget exhausted**

How Do We Validate Results?

Total Influence Spread

How many nodes are reached? Compare against traditional methods greedy (CELF) and sampling-based (IMM) methods

Spread-vs-Budget Curve

Plot reached nodes as budget increases. Does our method stay above baselines at every budget level?

Community Distribution

What fraction of each community is reached? Can we hit a balanced or targeted profile on demand?

Runtime Scalability

Wall-clock time on large graphs. Does RL scale where greedy methods become intractable?

Dataset (SNAP)

- **Stanford Large Network Dataset Collection**
- LiveJournal social network
  - Nodes: 3,997,962
  - Edges(Undirected): 34,681,189
  - Communities: 287,512

Baselines

- CELF — Cost Effective Lazy Forward selection
- IMM — Influence Maximization via Martingales